

Markscheme

Mathematics: analysis and approaches

Practice question bank

104. (a)

It can be proven by contradiction that $\frac{a+b}{2} > \sqrt{ab}$ for positive real numbers $a \neq b$ (assume $\frac{a+b}{2} \leq \sqrt{ab}$, square both sides, and show this forces $(a-b)^2 \leq 0$, a contradiction unless $a = b$).

Using this result, evaluate $\frac{a+b}{2} - \sqrt{ab}$ for $a = 9$ and $b = 4$.

$$\frac{9+4}{2} = 6.5 \quad (M1)$$

$$\sqrt{9 \times 4} = 6 \quad \text{A1}$$

$$6.5 - 6 = 0.5 \quad \text{A1}$$

[6 marks]